

Full Stack Web Development Lab

Week 8 – Record

Aim

To develop a mini web application, **Premium Fix** (a home services booking application), using HTML, CSS, JavaScript, Node.js and Express.js, with user registration, login, a dashboard and service booking, and JSON files used as a simple database.

Description

Premium Fix lets users book home services such as Plumber, Electrician, Home Salon, Home Makeup, Cleaning and Laundry. The **landing page** (index.html) introduces the application, shows the six services as cards and provides navigation to Home, Register and Login. The **registration page** collects name, age, date of birth, gender, mobile number, email, username, password and address, and sends them to the Express backend with a POST request; the server reads them from **req.body**, checks for duplicate users and appends the new user to **users.json** without deleting earlier users. The **login page** sends the username/email and password to a POST /login route, which validates them against users.json and returns a success or error message. After login, the **dashboard** shows a welcome message, the service options and all bookings of the user. Clicking a service opens a **booking page** where the user enters the date, time, address and notes; the details are sent to POST /book and stored in **bookings.json**, and the bookings of that service are listed below the form (GET /bookings). A **logout** option clears the session and returns to the login page. Flow: Landing Page → Registration → Store in JSON → Login → Validate Credentials → Dashboard → Book Service → Logout.

Program

app.js (Express Server)

```
const express = require('express');
const fs = require('fs');
const path = require('path');

const app = express();
app.use(express.json());
app.use(express.static(path.join(__dirname, 'public')));

// JSON files act as the database
const read = file => JSON.parse(fs.readFileSync(path.join(__dirname, file), 'utf-8'));
const write = (file, data) => fs.writeFileSync(path.join(__dirname, file), JSON.stringify(data, null, 2));
const page = name => (req, res) => res.sendFile(path.join(__dirname, 'public', name));

app.get('/', page('index.html'));
app.get('/register', page('register.html'));
app.get('/login', page('login.html'));

// Register: add new user to users.json
app.post('/register', (req, res) => {
  const { name, age, dob, gender, email, mobile, username, password, address } = req.body;
  if (!name || !email || !username || !password) {
    return res.status(400).json({ success: false, message: 'Please fill all required fields' });
  }
  const users = read('users.json');
  if (users.find(u => u.username === username || u.email === email)) {
    return res.status(409).json({ success: false, message: 'Username or email already registered' });
  }
  users.push({ name, age, dob, gender, email, mobile, username, password, address });
  write('users.json', users);
  res.json({ success: true, message: 'Registration successful! Redirecting to login...' });
});

// Login: check username/email + password
app.post('/login', (req, res) => {
  const { username, password } = req.body;
  const user = read('users.json').find(
    u => (u.username === username || u.email === username) && u.password === password
  );
  if (!user) {
    return res.status(401).json({ success: false, message: 'Invalid credentials' });
  }
  // Here you would typically set a session or cookie for the user
  res.json({ success: true, message: 'Login successful!' });
});
```

```

    });
    if (!user) return res.status(401).json({ success: false, message: 'Invalid username or password' });
    res.json({ success: true, user: { name: user.name, username: user.username } });
  });

// Book a service: add to bookings.json
app.post('/book', (req, res) => {
  const { username, service, date, time, address, notes } = req.body;
  if (!username || !service || !date || !time || !address) {
    return res.status(400).json({ success: false, message: 'Please fill all required fields' });
  }
  const bookings = read('bookings.json');
  bookings.push({ id: Date.now(), username, service, date, time, address, notes: notes || '', status: 'Booked' });
  write('bookings.json', bookings);
  res.json({ success: true, message: 'Booking confirmed!' });
});

// List bookings of a user (optionally only one service)
app.get('/bookings', (req, res) => {
  const { username, service } = req.query;
  res.json(read('bookings.json').filter(b => b.username === username && (!service || b.service === service)));
});

app.listen(3000, () => console.log('Premium Fix running at http://localhost:3000'));

```

package.json

```

{
  "name": "premium-fix",
  "version": "1.0.0",
  "main": "app.js",
  "scripts": { "start": "node app.js" },
  "dependencies": { "express": "^5.2.1" }
}

```

users.json (Users Database)

```

[]

```

bookings.json (Bookings Database)

```

[]

```

public/index.html (Landing Page)

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Premium Fix</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <div class="navbar" id="nav"></div>
  <div class="hero">
    <h1>Premium Fix</h1>
    <p>Trusted home services at your doorstep. Book in seconds.</p>
    <a class="btn" href="register.html">Get Started</a>
    <a class="btn alt" href="login.html">Login</a>
  </div>
  <div class="container">
    <h2 class="section">Our Services</h2>
    <div class="grid" id="services"></div>
  </div>
  <footer>&copy; 2026 Premium Fix | Home Services</footer>
  <script src="js/script.js"></script>
</body>
</html>

```

public/register.html (Registration Page)

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Premium Fix</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <div class="navbar" id="nav"></div>
  <div class="container">
    <div class="panel wide">

```

```

<h2>Create Account</h2>
<form id="registerForm">
  <label>Full Name</label><input name="name" required>
  <div class="row">
    <div><label>Age</label><input type="number" name="age" min="1" required></div>
    <div><label>Date of Birth</label><input type="date" name="dob" required></div>
  </div>
  <div class="row">
    <div><label>Gender</label>
    <select name="gender" required><option value="">Select</option><option>Male</option><option>Female</option><option>Other</option></select></div>
    <div><label>Mobile Number</label><input name="mobile" pattern="[0-9]{10}" title="10 digit number" required></div>
  </div>
  <label>Email</label><input type="email" name="email" required>
  <div class="row">
    <div><label>Username</label><input name="username" required></div>
    <div><label>Password</label><input type="password" name="password" required></div>
  </div>
  <label>Address</label><textarea name="address" required></textarea>
  <button class="btn" type="submit">Register</button>
</form>
<p id="msg"></p>
<p>Already have an account? <a href="login.html">Login</a></p>
</div>
</div>
<script src="js/script.js"></script>
</body>
</html>

```

public/login.html (Login Page)

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Premium Fix</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <div class="navbar" id="nav"></div>
  <div class="container">
    <div class="panel">
      <h2>Login</h2>
      <form id="loginForm">
        <label>Username or Email</label><input name="username" required>
        <label>Password</label><input type="password" name="password" required>
        <button class="btn" type="submit">Login</button>
      </form>
      <p id="msg"></p>
      <p>New here? <a href="register.html">Register</a></p>
    </div>
  </div>
  <script src="js/script.js"></script>
</body>
</html>

```

public/dashboard.html (Dashboard)

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Premium Fix</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <div class="navbar" id="nav"></div>
  <div class="container">
    <h2 id="welcome"></h2>
    <h3>Book a Service</h3>
    <div class="grid" id="services"></div>
    <h3 style="margin-top:35px">My Bookings</h3>
    <div class="list" id="list" style="margin:0"></div>
  </div>
  <script src="js/script.js"></script>
</body>
</html>

```

public/service.html (Service Booking Page)

```

<!DOCTYPE html>
<html>

```

```

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Premium Fix</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <div class="navbar" id="nav"></div>
  <div class="container">
    <div class="panel wide">
      <h2 id="title"></h2>
      <p id="desc"></p>
      <form id="bookForm">
        <div class="row">
          <div><label>Date</label><input type="date" name="date" id="date" required></div>
          <div><label>Time</label><input type="time" name="time" required></div>
        </div>
        <label>Address</label><input name="address" required>
        <label>Notes (optional)</label><textarea name="notes"></textarea>
        <button class="btn" type="submit">Book Now</button>
      </form>
      <p id="msg"></p>
    </div>
    <h3 style="text-align:center">Your Bookings</h3>
    <div class="list" id="list"></div>
  </div>
  <script src="js/script.js"></script>
</body>
</html>

```

public/css/style.css

```

* { box-sizing: border-box; }
body { margin: 0; font-family: 'Segoe UI', Arial, sans-serif; background: #f5f6fa; color: #222; }
.navbar { background: #0f172a; display: flex; justify-content: space-between; align-items: center; padding: 14px 40px; }
.navbar a { color: #e2e8f0; text-decoration: none; margin-left: 22px; }
.navbar .brand { font-size: 22px; font-weight: bold; margin: 0; color: white; }
.brand span { color: #f59e0b; }
.hero { background: linear-gradient(135deg, #0f172a, #1e3a8a); color: white; text-align: center; padding: 70px 20px; }
.hero h1 { font-size: 40px; margin: 0 0 10px; }
.hero p { color: #cbd5e1; font-size: 18px; margin-bottom: 25px; }
.btn { display: inline-block; background: #f59e0b; color: #111; padding: 12px 26px; border-radius: 6px; text-decoration: none; font-weight: bold; border: none; cursor: pointer; font-size: 15px; margin: 6px; }
.btn.alt { background: transparent; color: white; border: 1px solid white; }
.container { max-width: 1000px; margin: 35px auto; padding: 0 20px; }
h2.section { text-align: center; margin-bottom: 25px; }
.grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(150px, 1fr)); gap: 18px; }
.service { background: white; border-radius: 12px; padding: 25px 10px; text-align: center; text-decoration: none; color: #222; box-shadow: 0 2px 8px rgba(0,0,0,0.08); transition: 0.2s; }
.service:hover { transform: translateY(-5px); box-shadow: 0 8px 18px rgba(0,0,0,0.15); }
.service .icon { font-size: 38px; }
.service h4 { margin: 10px 0 4px; }
.service p { margin: 0; font-size: 13px; color: #666; padding: 0 8px; }
.panel { background: white; max-width: 460px; margin: 0 auto 25px; padding: 28px; border-radius: 12px; box-shadow: 0 2px 10px rgba(0,0,0,0.1); }
.panel.wide { max-width: 600px; }
.panel h2 { margin-top: 0; }
label { display: block; margin-top: 12px; font-size: 14px; font-weight: 600; }
input, select, textarea { width: 100%; padding: 10px; margin-top: 5px; border: 1px solid #ccc; border-radius: 6px; font-size: 14px; font-family: inherit; }
form .btn { width: 100%; margin: 18px 0 0; }
.row { display: grid; grid-template-columns: 1fr 1fr; gap: 12px; }
.msg { margin-top: 12px; font-size: 14px; }
.error { color: #dc2626; }
.ok { color: #16a34a; }
.booking { background: white; border-left: 5px solid #f59e0b; border-radius: 8px; padding: 14px 18px; margin-bottom: 12px; box-shadow: 0 1px 5px rgba(0,0,0,0.08); line-height: 1.6; }
.tag { float: right; background: #dcfce7; color: #166534; padding: 3px 10px; border-radius: 20px; font-size: 12px; }
.list { max-width: 600px; margin: 0 auto; }
footer { text-align: center; padding: 20px; color: #777; font-size: 13px; }

```

public/js/script.js

```

const SERVICES = [
  { id: 'plumber', name: 'Plumber', icon: '■', desc: 'Leaks, taps, pipes & fittings' },
  { id: 'electrician', name: 'Electrician', icon: '■', desc: 'Wiring, switches, fans & lights' },
  { id: 'salon', name: 'Home Salon', icon: '■', desc: 'Haircut, facial & grooming at home' },
  { id: 'makeup', name: 'Home Makeup', icon: '■', desc: 'Party & bridal makeup artists' },
  { id: 'cleaning', name: 'Cleaning', icon: '■', desc: 'Deep cleaning for home & kitchen' },
  { id: 'laundry', name: 'Laundry', icon: '■', desc: 'Wash, dry & iron pickup service' }
];
const user = JSON.parse(localStorage.getItem('user') || 'null');

```

```

const esc = s => String(s).replace(/[\<>"]/g, c => ({ '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;' }[c]));
const serviceName = id => (SERVICES.find(s => s.id === id) || { name: id }).name;

async function post(url, data) {
  const r = await fetch(url, { method: 'POST', headers: { 'Content-Type': 'application/json' }, body: JSON.stringify(data) });
  return r.json();
}

function showMsg(r) {
  const msg = document.getElementById('msg');
  msg.className = 'msg ' + (r.success ? 'ok' : 'error');
  msg.innerText = r.message;
}

function logout() { localStorage.removeItem('user'); location.href = 'login.html'; }
function requireLogin() { if (!user) { location.href = 'login.html'; return false; } return true; }

function renderNav() {
  const links = user
    ? '<a href="dashboard.html">Dashboard</a><a href="#" onclick="logout()">Logout</a>'
    : '<a href="register.html">Register</a><a href="login.html">Login</a>';
  document.getElementById('nav').innerHTML =
    '<a class="brand" href="index.html">Premium<span>Fix</span></a><div><a href="index.html">Home</a>' + links +
    '</div>';
}

function showServices(id) {
  document.getElementById(id).innerHTML = SERVICES.map(s =>
    '<a class="service" href="service.html?type=' + s.id + '"><div class="icon">' + s.icon + '</div><h4>' + s.name + '</h4><p>' + s.desc + '</p></a>'
  ).join('');
}

async function loadBookings(id, service) {
  const url = '/bookings?username=' + encodeURIComponent(user.username) + (service ? '&service=' + service : '');
  const list = (await (await fetch(url)).json()).reverse();
  document.getElementById(id).innerHTML = list.length ? list.map(b =>
    '<div class="booking"><span class="tag">' + esc(b.status) + '</span><b>' + esc(serviceName(b.service)) + '</b><br>' +
    esc(b.date) + ' at ' + esc(b.time) + '<br>' + esc(b.address) + (b.notes ? '<br><i>' + esc(b.notes) + '</i>' : '') + '</div>'
  ).join('') : '<p>No bookings yet.</p>';
}

renderNav();
if (document.getElementById('services')) showServices('services');

// Register page
const regForm = document.getElementById('registerForm');
if (regForm) regForm.addEventListener('submit', async e => {
  e.preventDefault();
  const r = await post('/register', Object.fromEntries(new FormData(regForm)));
  showMsg(r);
  if (r.success) setTimeout(() => location.href = 'login.html', 1200);
});

// Login page
const loginForm = document.getElementById('loginForm');
if (loginForm) loginForm.addEventListener('submit', async e => {
  e.preventDefault();
  const r = await post('/login', Object.fromEntries(new FormData(loginForm)));
  if (r.success) {
    localStorage.setItem('user', JSON.stringify(r.user));
    location.href = 'dashboard.html';
  } else showMsg(r);
});

// Dashboard page
if (document.getElementById('welcome') && requireLogin()) {
  document.getElementById('welcome').innerText = 'Welcome, ' + user.name + '!';
  loadBookings('list');
}

// Service booking page
const bookForm = document.getElementById('bookForm');
if (bookForm && requireLogin()) {
  const svc = SERVICES.find(s => s.id === new URLSearchParams(location.search).get('type')) || SERVICES[0];
  document.getElementById('title').innerText = svc.icon + ' Book ' + svc.name;
  document.getElementById('desc').innerText = svc.desc;
  document.getElementById('date').min = new Date().toISOString().split('T')[0];
  loadBookings('list', svc.id);
  bookForm.addEventListener('submit', async e => {

```

```

e.preventDefault();
const data = Object.fromEntries(new FormData(bookForm));
const r = await post('/book', { ...data, username: user.username, service: svc.id });
showMsg(r);
if (r.success) { bookForm.reset(); loadBookings('list', svc.id); }
});
}

```

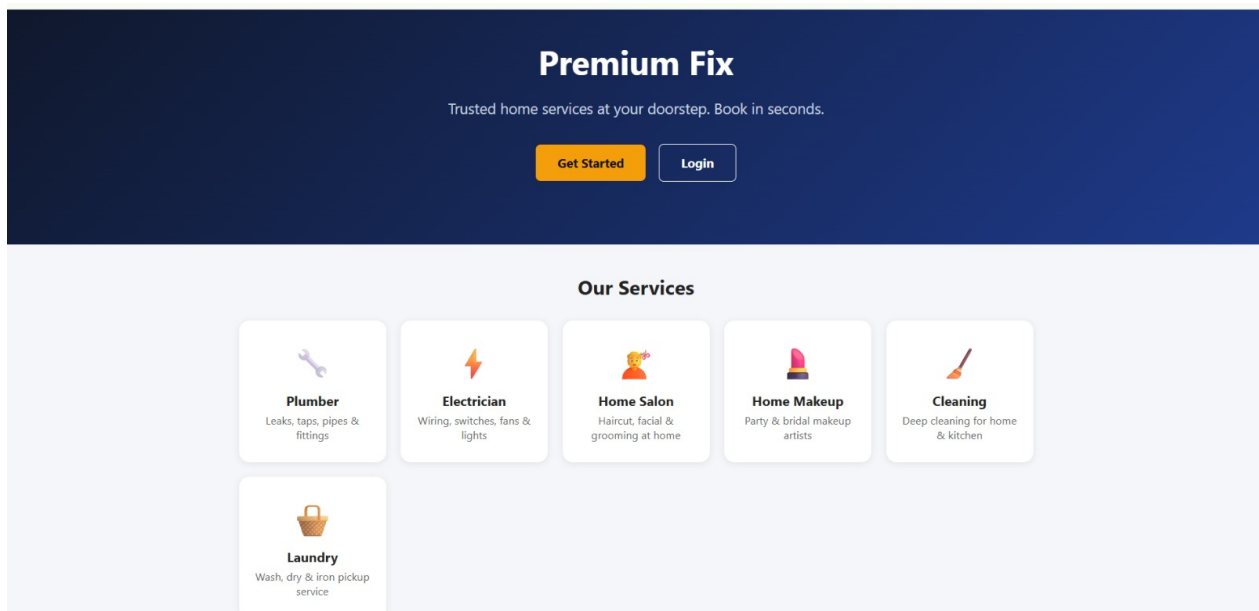
Details about the Program

Where Saved	All files are saved inside the project folder FL_LAB_ANITS/fs_week_8/ in the GitHub repository raviteja-codehub/FL_LAB_ANITS — app.js (server), users.json and bookings.json (JSON database), package.json, and the public/ folder with the HTML pages, css/style.css and js/script.js. node_modules is not uploaded.
Where Executed	The server runs on Node.js from the terminal on the local computer, and the application is opened in a web browser (Google Chrome / Edge / Firefox) at http://localhost:3000. GitHub Pages cannot run Node.js, so this week is not hosted online.
How Executed	1. Open a terminal in the fs_week_8 folder. 2. Run npm install to install Express. 3. Start the server with node app.js; the console prints "Premium Fix running at http://localhost:3000". 4. Open http://localhost:3000 in the browser, register a new user, log in, choose a service, book it and view the booking list. Use Logout to return to the login page.

Output

After starting the server with node app.js and opening http://localhost:3000, the landing page, registration page and service booking page appear as shown below:

Landing Page (services list)



Registration Page

Create Account

Full Name

Raviteja Pasupuleti

Age

20

Date of Birth

22-08-2006

Gender

Male

Mobile Number

8886898943

Email

ravi@123

Username

ravi@123

Password

.....

Address

vizag,anits rod tagarapuvalasa,2/43

Register

Already have an account? [Login](#)

Service Booking Page (Electrician)

Book Electrician

Wiring, switches, fans & lights

Date

25-10-2026

Time

10:00

Address

2/43,anits rroad,vizag

Notes (optional)

come on time

Book Now

Your Bookings

No bookings yet.